

Inverse Problems in Geophysics

Part 1: Introduction

MGPY, MGIN, MDR, MGEX-CMG, PhD

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Introduction

- Content and dates
- Literature and Links
- definition of data and model
- error and noise
- linear vs. non-linear
- linear inversion: simple matrix problem

Subject and Objectives

! Inversion (main task in applied geophysics)

Determine a **model** describing the subsurface that can **explain our data!**

i Aim of the lecture

- understand the process of imaging/inversion
- solve inverse problems by simple scripts
- actively control the inversion process
- use tools to get a feeling for the value of models

Content

0. Introduction, data, models
1. Simple matrix problems, linear regression
2. Method of least squares
3. Resolution and Singular Value Decomposition
4. Regularization methods
5. Ray tomography
6. Non-linear minimization
7. Time-lapse inversion
8. Global optimization methods

Schedule

Lectures Wednesday, 11:30-13, MEI-0150

14 slots: 8.4., 15.4., 22.4., 29.4., 6.5., 13.5., 20.5., 27.5., 3.6., 10.6., 17.6., 1.7., 8.7., 15.7.

Exercises Friday, 11:30-13, CIP pool MEI1203a (or 1203b)

14 slots: 7.4., 14.4., 21.4., 28.4., 5.5., 12.5., 19.5., 26.5., 2.6., 9.6., 16.6., 23.6., 30.6., 7.7., 14.7.

Grade: submitting two reports including codes (Notebooks)

What should you know already?

- Higher mathematics: differential equations, algebra (1.-2. BGIP)
- Experimental and theoretical physics: governing equations
- Numerics for engineers (2. BGIP)
- Programming (1. BGIP), Software development (3. BGIP)
- Geophysics: feeling for physical fields & methods (2.-4. BGIP)
- Electromagnetics (5. BGIP), Theory EM,
- MSc level: Scientific programming, HPC, seismic imaging

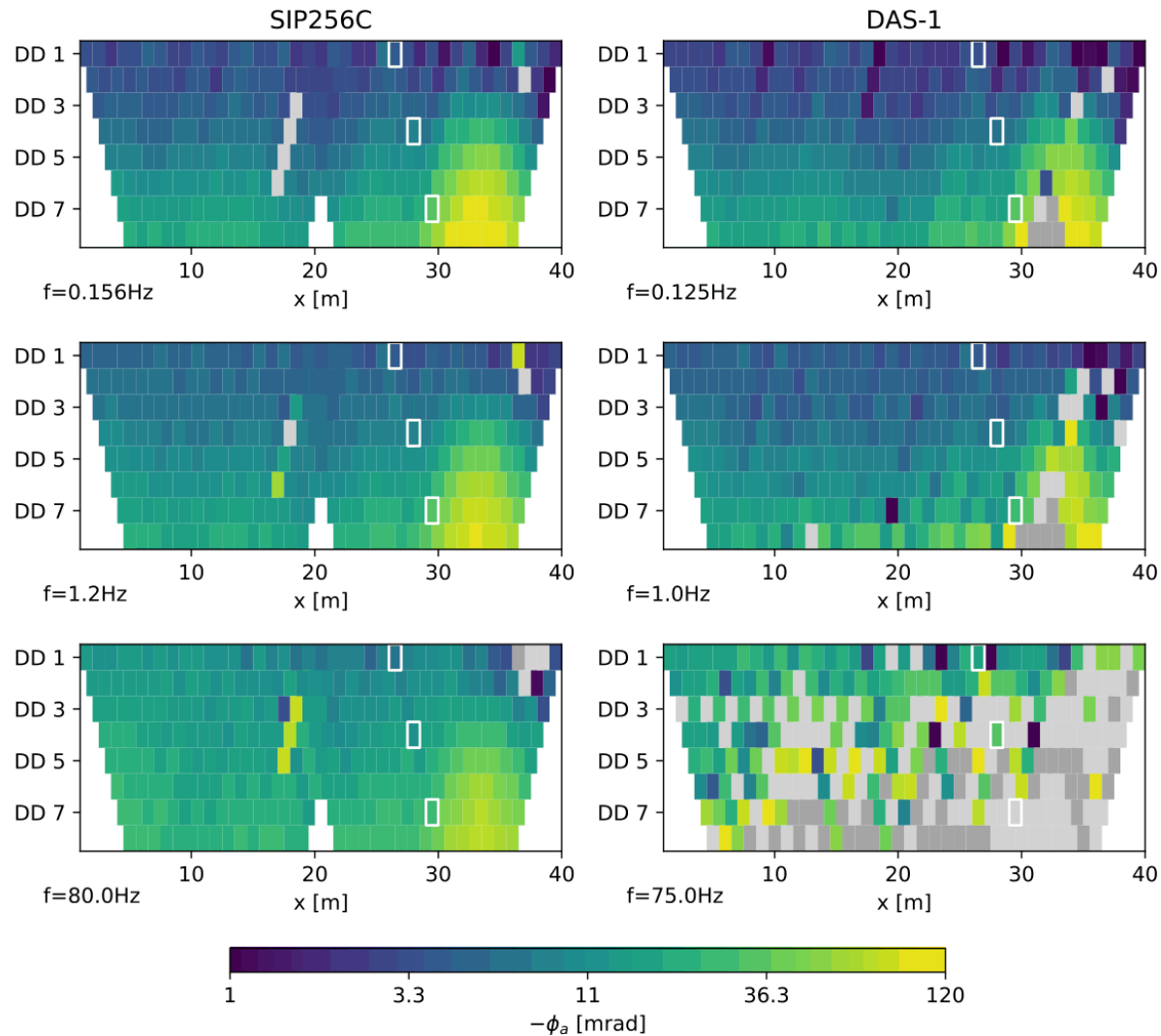
Literature

- Menke (2018): [Geophysical Data Analysis: Discrete Inverse Theory](#), Academic
- Gubbins, Tarantola, Zhdanov
- Richter (2020): [Inverse Problems: Basics, Theory and Applications in Geophysics](#)
- Günther (2004): [Inversion Methods and Resolution Analysis for the 2D/3D Reconstruction of Resistivity Structures from DC Measurements](#), PhD thesis

Further links

- pyGIMLi: Python Geophysical Inversion and Modelling Library
<https://pygimli.org>
- Geoscience.XYZ: <https://geosci.xyz>
- <https://github.com/halbmy/IJulia> - Julia Notebooks for Inverse Problems

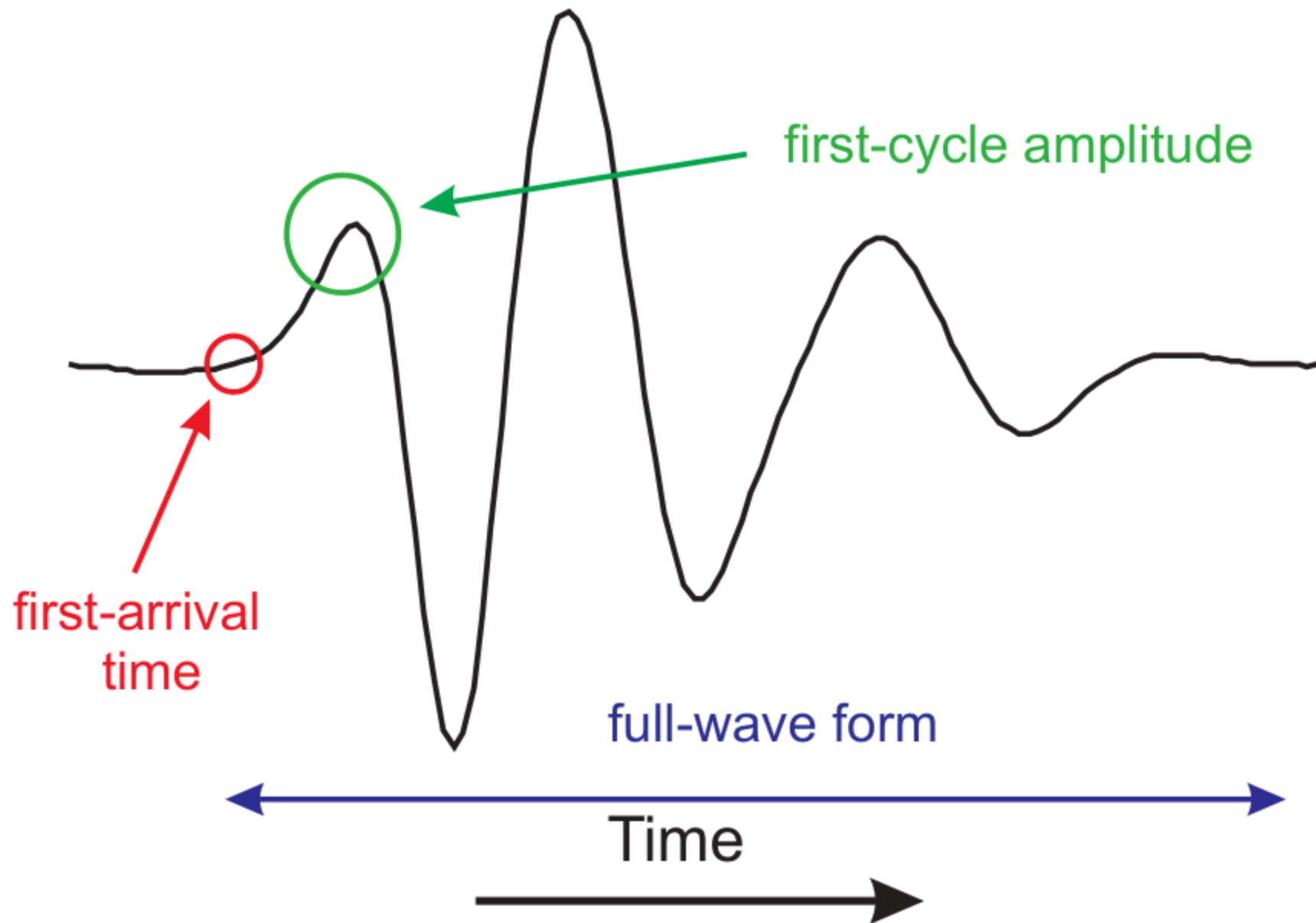
Data



- left: ERT/IP pseudosections for different frequencies
- single values
- often arranged logically view on data important
- sometimes associated to space

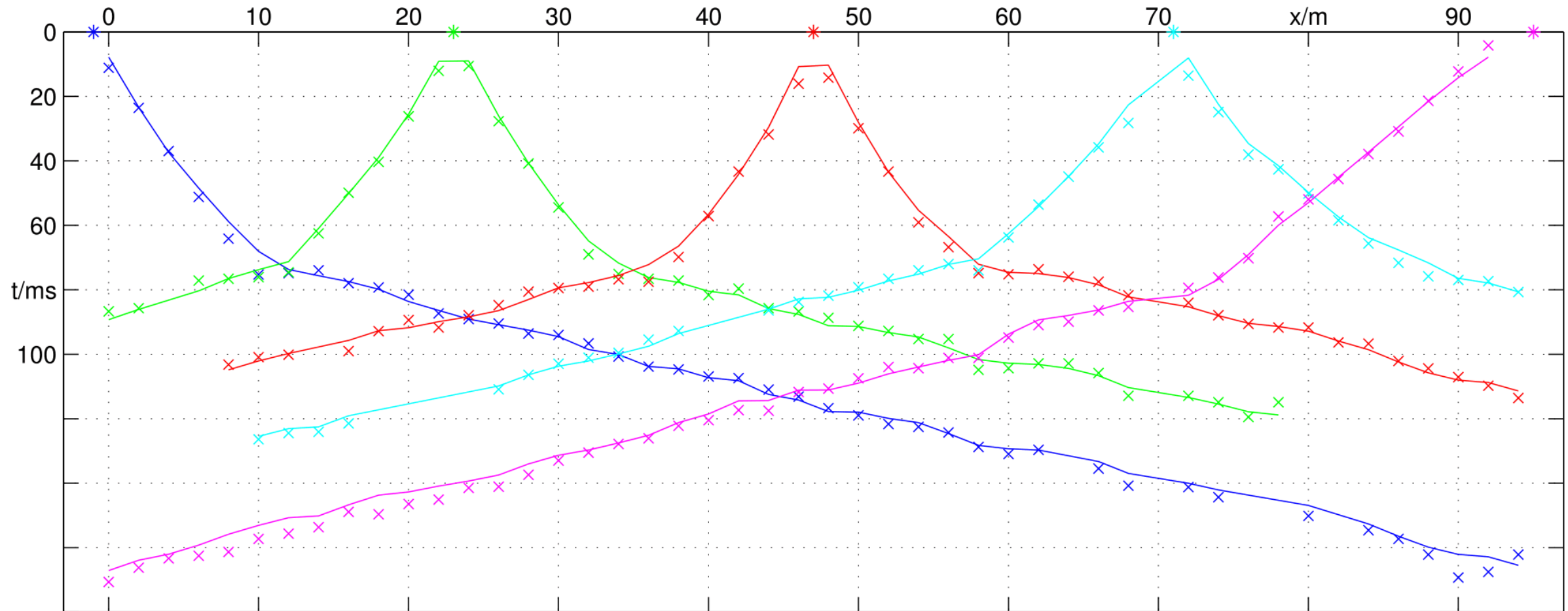
Data in seismology

time series of acceleration, velocity or position

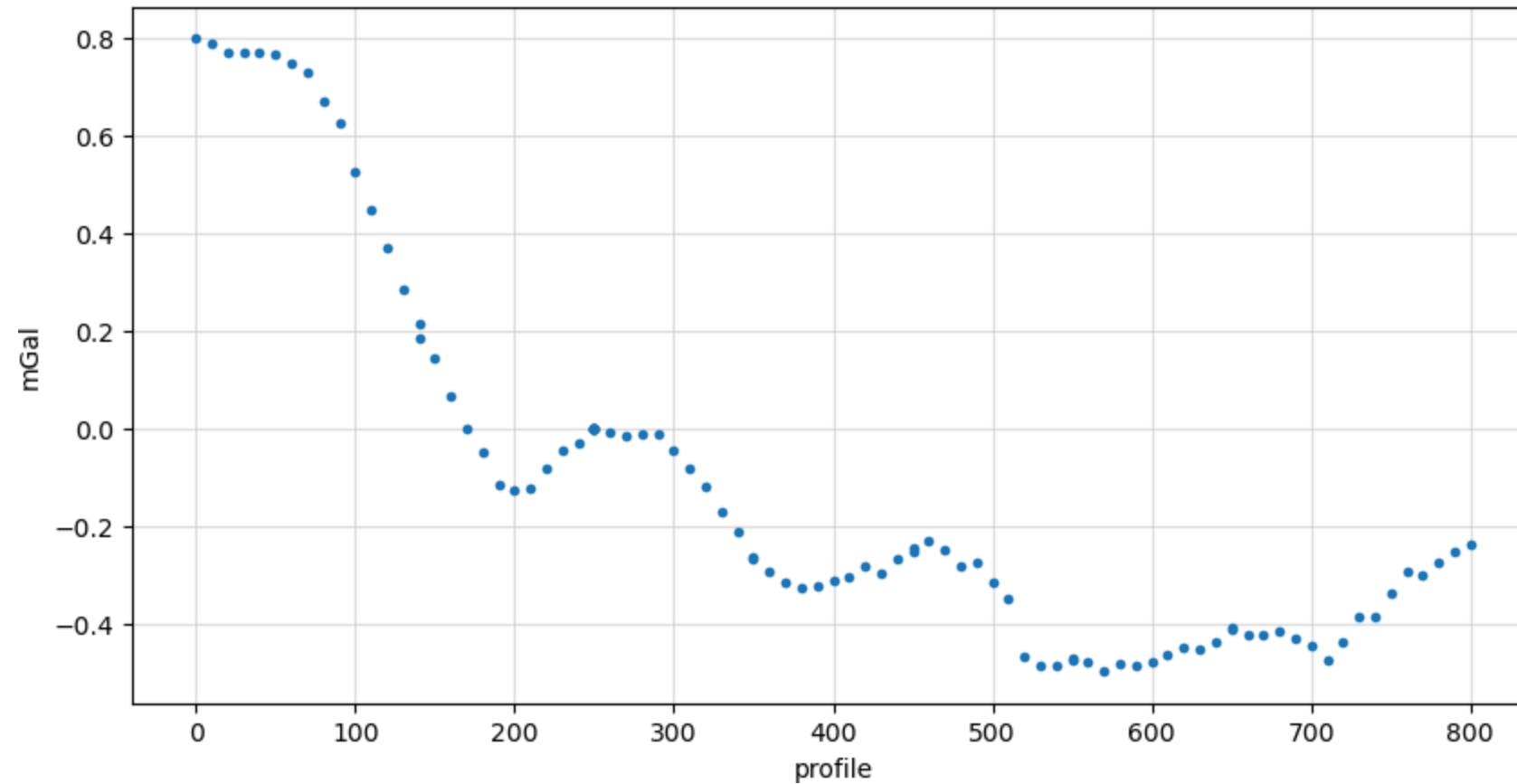


Data in travel-time tomography

traveltimes between shot and geophones



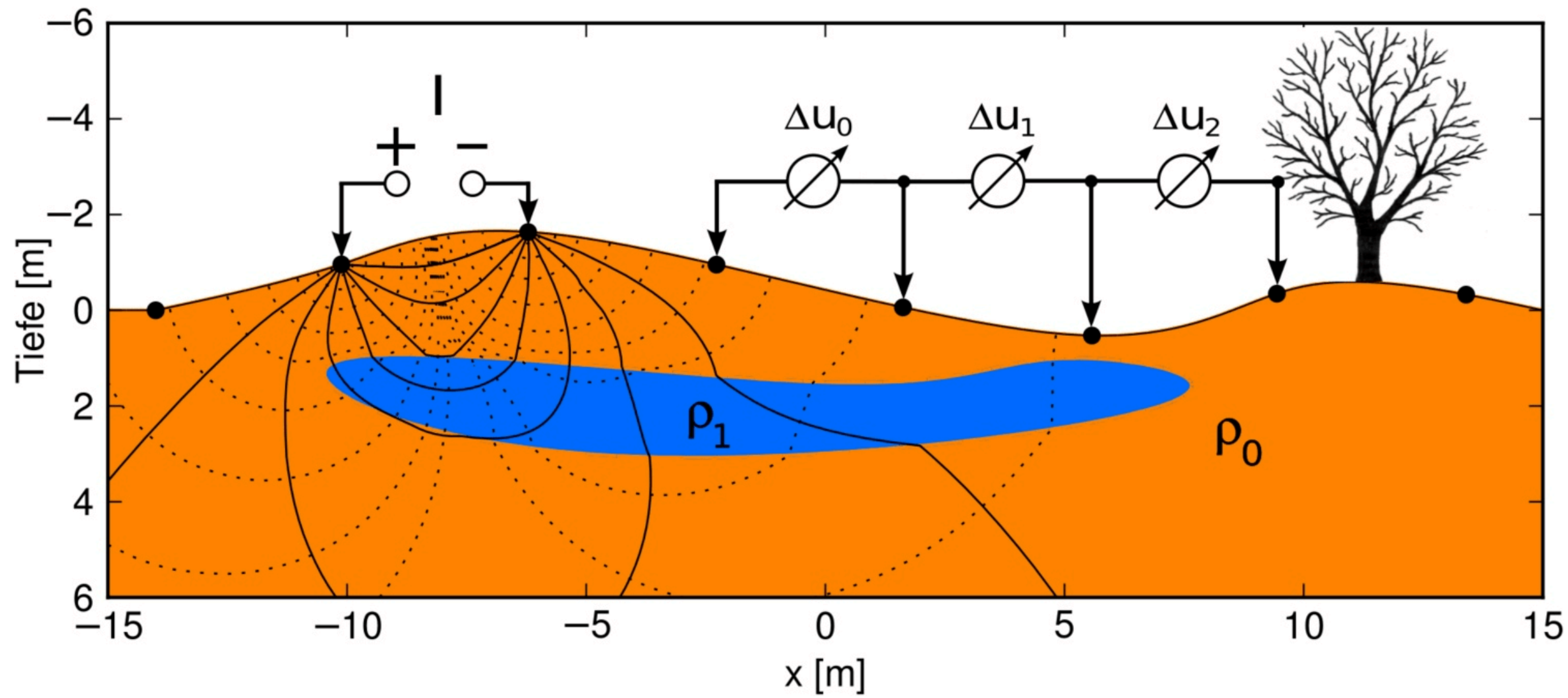
Data in gravimetry



Bouguer anomaly of gravity profile in Zellwald

Data in electrical resistivity tomography (ERT)

current and voltage combinations



Data

- can be a discretized function of space, time or frequency, and plotted as such (curves)
- can depend on several sensor (shot/geophone, ABMN) and visualized as several curves or a coloured matrix (crossplot, pseudosection)

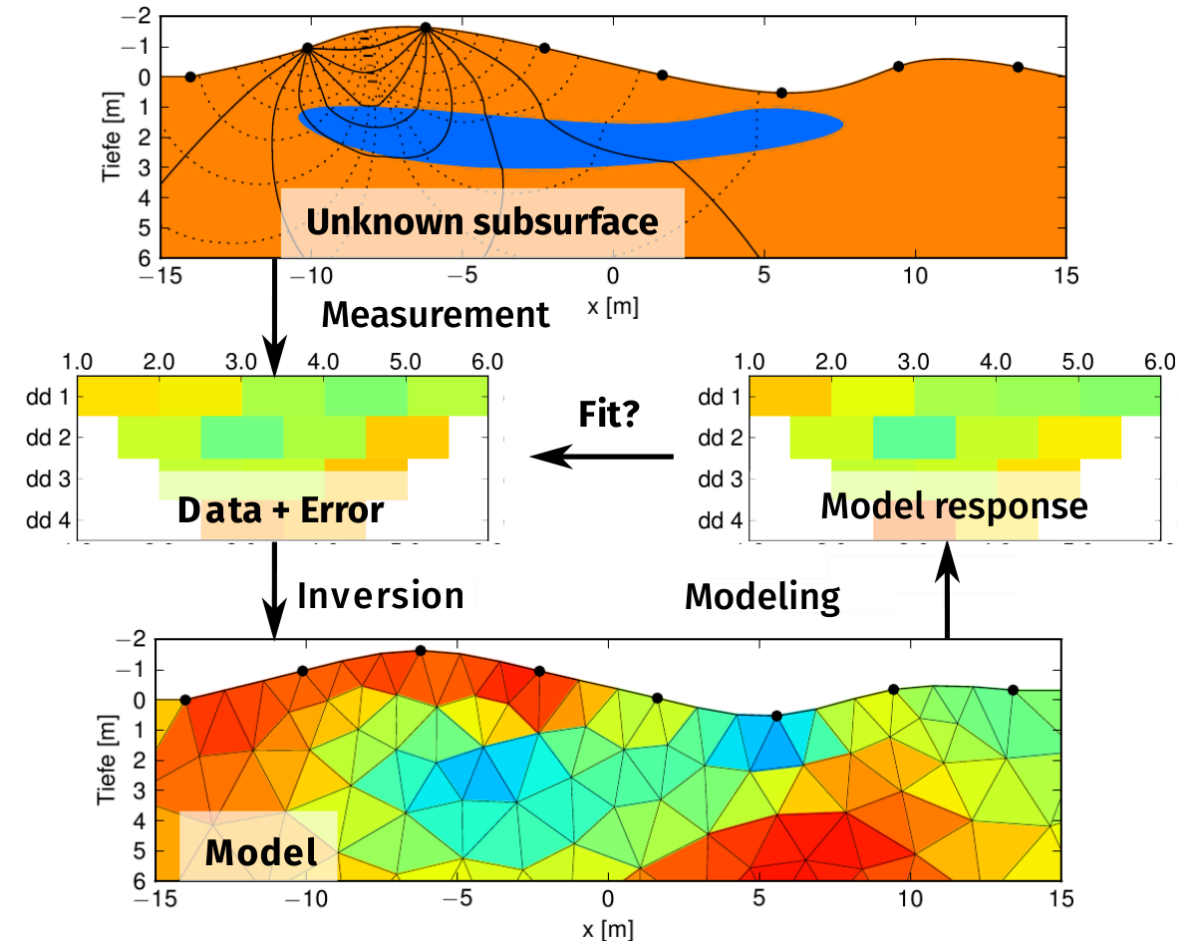
Data are subsumed in the *data vector*

$$\mathbf{d} = [d_1, d_2, \dots, d_N]^T$$

consist of true *model response* $\mathbf{f}(\mathbf{m})$ plus *noise* $\mathbf{n}$: $\mathbf{d} = \mathbf{f}(\mathbf{m}) + \mathbf{n}$

Geophysical Workflow

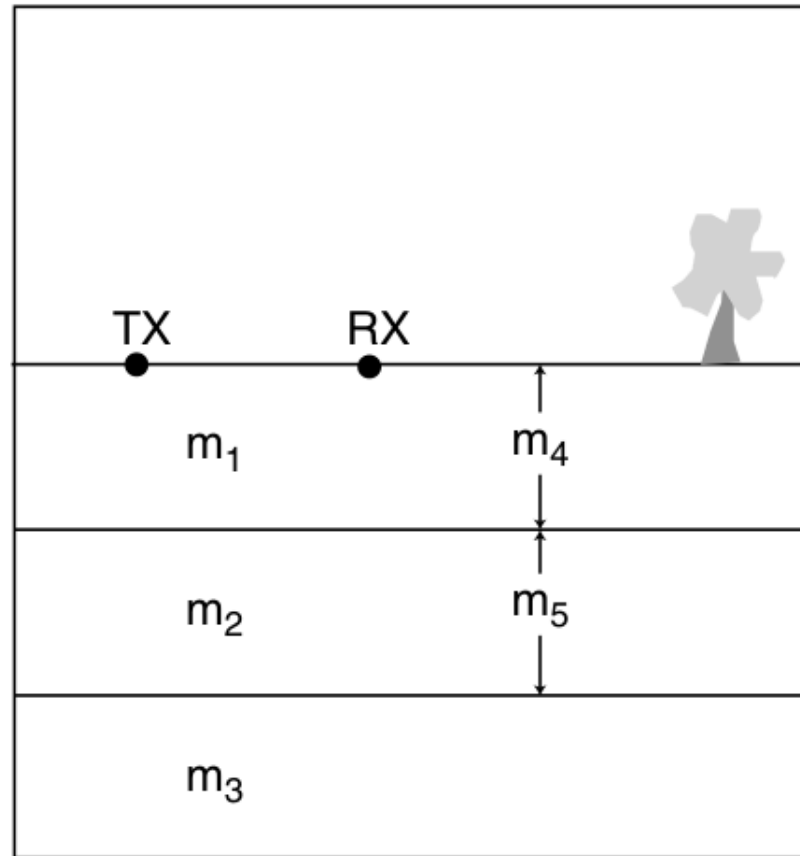
1. Data acquisition
2. Preprocessing (QC, filtering)
3. Parameterization (mesh)
4. **Inversion**
5. Fit of data & model response
6. Postprocessing & visualization
7. Interpretation



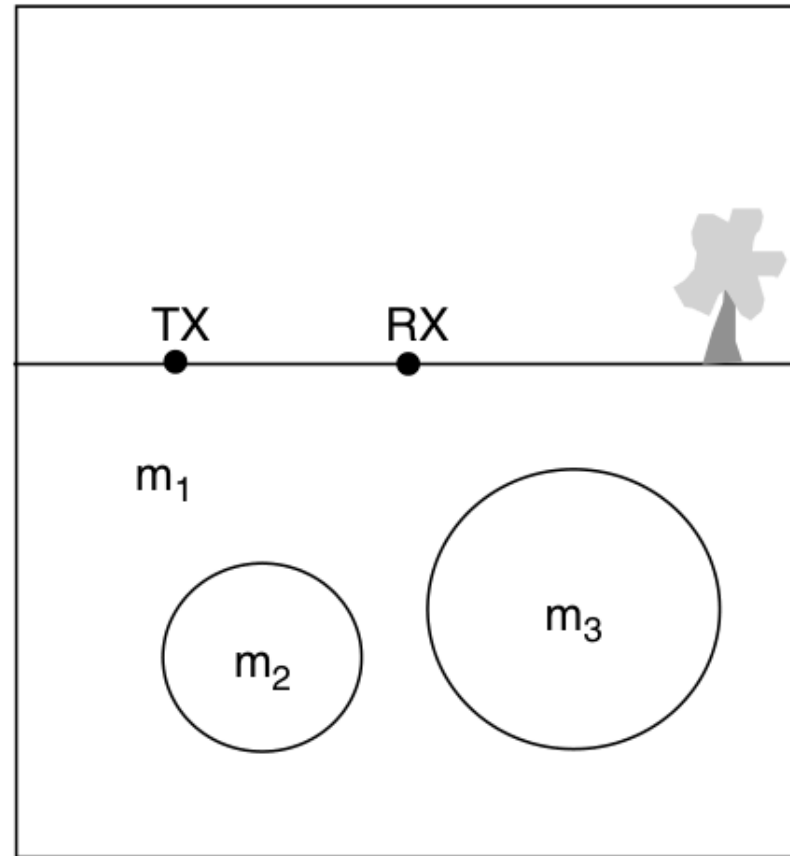
Model

- numerical parameterization of subsurface (our assumption)

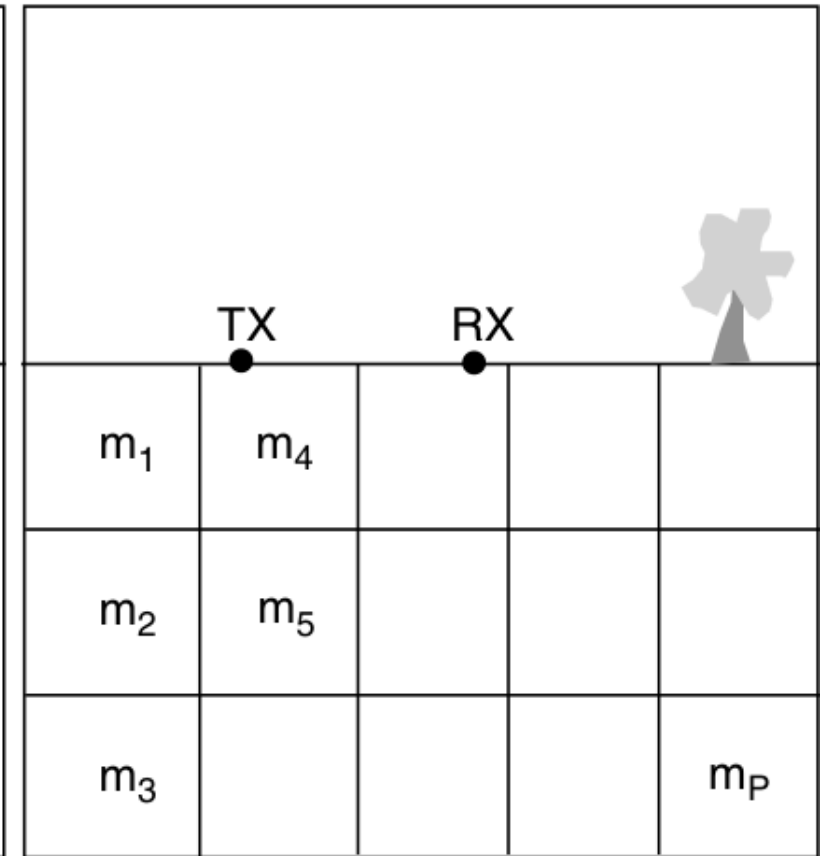
(a) layered model



(b) parametric model



(c) gridded model



Model types

subsurface described by *model vector* $\mathbf{m} = [m_1, m_2, \dots, m_M]^T$

Independent parameters:

- seismology: earthquake location, stress, principal axis
- gravity: position and depth of anomaly, size, density contrast
- spectroscopy (e.g. SIP): function parameters (e.g. Cole-Cole)
- layered model thicknesses and values (ρ , v)

Model types

subsurface described by *model vector* $\mathbf{m} = [m_1, m_2, \dots, m_M]^T$

Discrete functions of space, time, frequency

(same property aligned along axes)

- refraction: depth (and velocity) of refractor
- distribution of a parameter in space (and time)
- spectroscopy: Fourier or Debye distribution
- e.g. 2D/3D resistivity/velocity/density distribution

Inversion

i Inverse problem: determine $\mathbf{m}$ so that data are explained

$$\mathbf{d} \approx \mathbf{f}(\mathbf{m})$$

i Why not

$$\mathbf{d} = \mathbf{f}(\mathbf{m})$$

Because $\mathbf{d}$ contains random noise not to be explained.

Minimization problem

i Minimization problem

$$\|\mathbf{d} - \mathbf{f}(\mathbf{m})\|_2^2 = \sum_i^N (d_i - f_i(\mathbf{m}))^2 \rightarrow \min$$

Take into account errors: Explain model within error (ϵ) bounds

$$\left\| \frac{\mathbf{d} - \mathbf{f}(\mathbf{m})}{\epsilon} \right\| = \sum_i^N \left(\frac{d_i - f_i(\mathbf{m})}{\epsilon_i} \right)^2 \rightarrow \min$$

error model $\epsilon = [\epsilon_1, \epsilon_2, \dots, \epsilon_N]^T$ (assumption of noise standard deviation)

Correctness

Well-posed problems (Hadamard)

- There is a solution,
- it is uniquely defined and
- depends steadily from input data (small variations lead to small model deviations)

Ill-posed problem

- There is no model to fit the data perfectly
- Many models can fit the data within errors
- Small data variations can lead to large model deviations

Occam's razor - a fundamental rule

❗ novacula Occami

Pluralitas non est ponenda sine neccesitate!

(William of Ockham, Scottish philosopher and theologian, 14th century)

i Principle of Parsimony

Entities must not be multiplied beyond necessity.

Of two competing theories, the simpler explanation of an entity is to be preferred.

Linear problems

$\mathbf{f}(\mathbf{m})$ is linear with respect to $\mathbf{m} \Rightarrow$ write as matrix-vector product

$$\mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n}$$

Examples

Gravity, Magnetism, Magnetic Resonance, VSP, straight-ray tomography

Linear problems

$\mathbf{f}(\mathbf{m})$ is linear with respect to $\mathbf{m} \Rightarrow$ write as matrix-vector product

$$\mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n}$$

Problem

$\mathbf{m} = \mathbf{G}^{-1}\mathbf{d}$? No, because is usually not invertible, not even quadratic.

Types of inverse problems

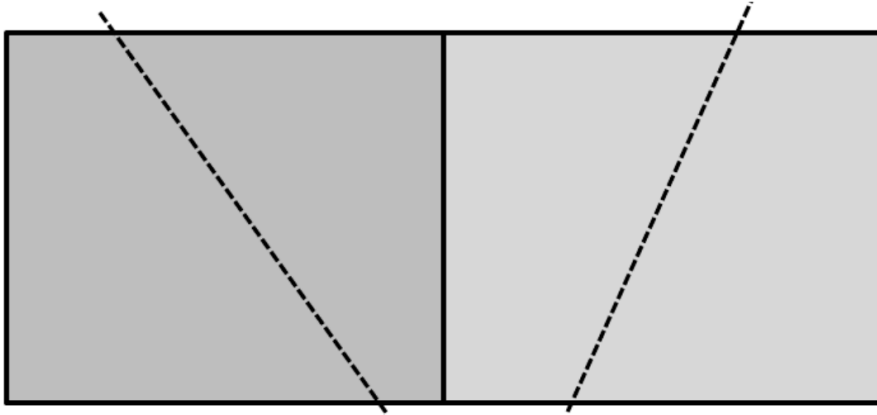
- ! • every row stands for a measurement (data point)
- every column represents a model parameter

$$\mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n} \Rightarrow \mathbf{G} \in \mathfrak{R}^{N \times M}$$

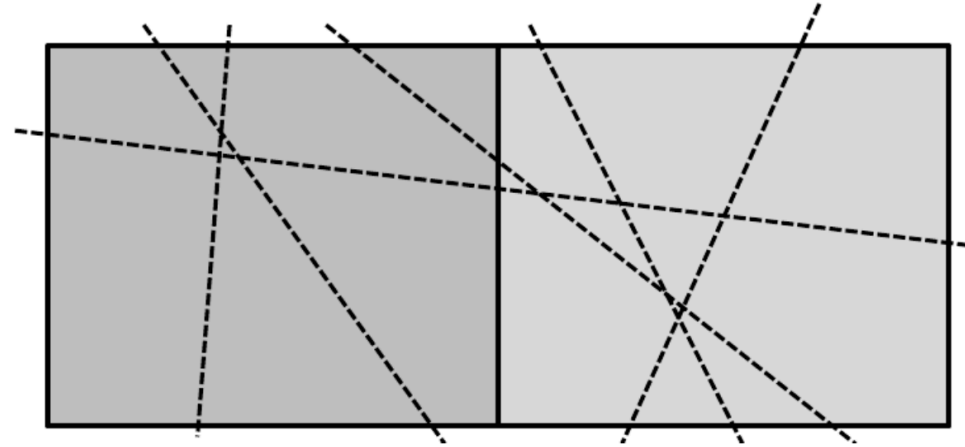
- $N = M$: even-determined
- $N > M$: over-determined problem (no existing solution)
- $N < M$: under-determined problem (no unique solution)
- mostly: (both over- and) under-determined model parts

Types of inverse problems (Menke, 2012)

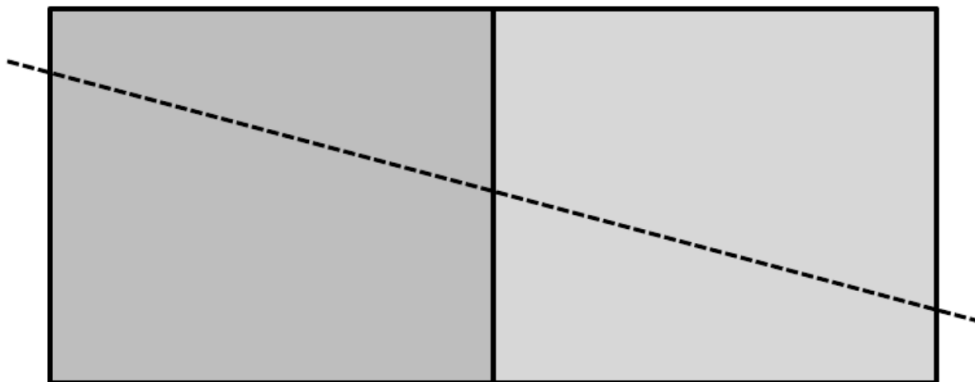
Even-determined



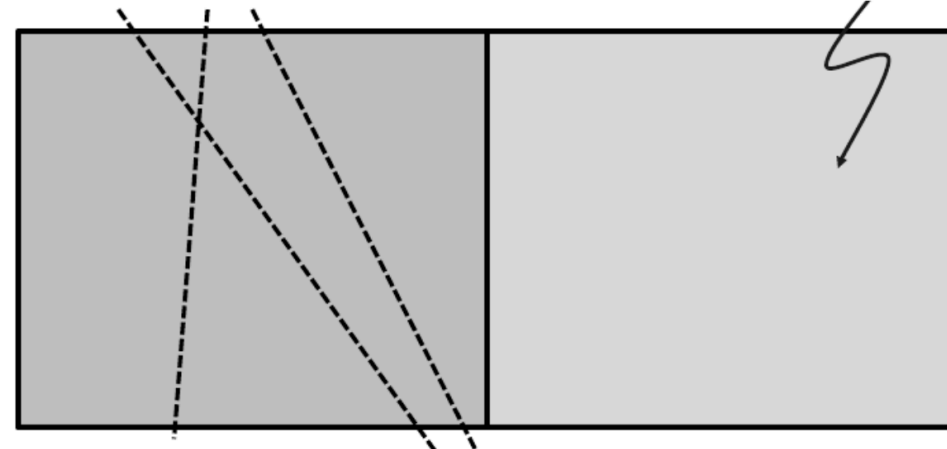
Over-determined



Under-determined



Mixed-determined



null space

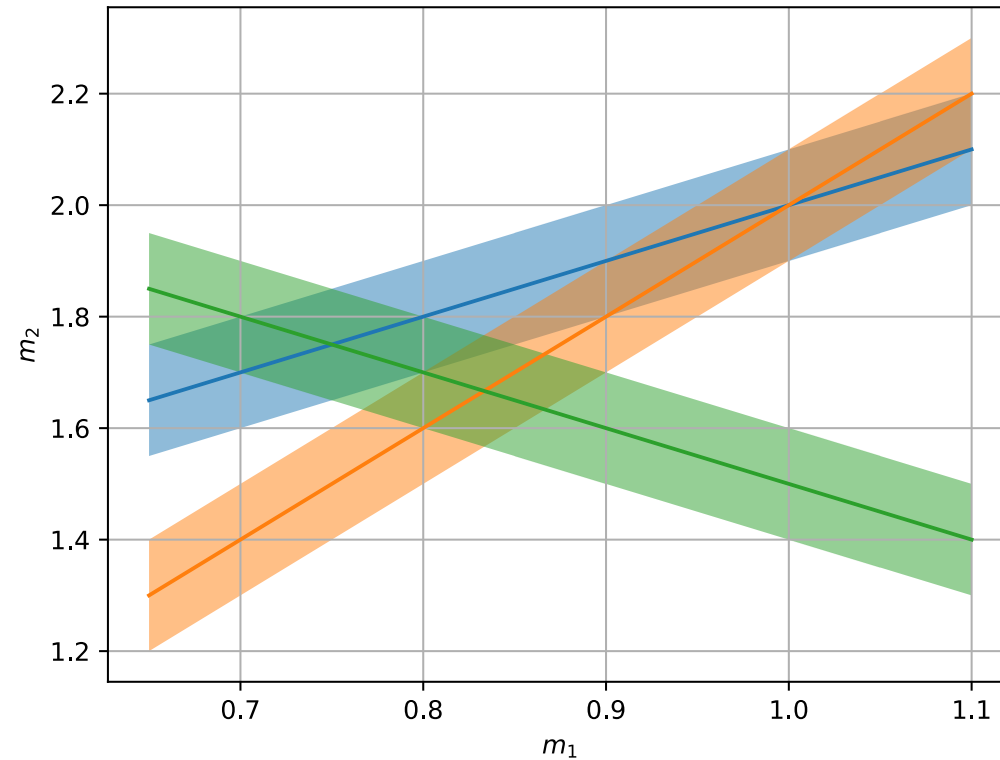
A simple matrix problem

$$m_1 - m_2 = -1$$

$$2m_1 - m_2 = 0$$

$$m_1 + m_2 = 2.5$$

$$m_2 = \frac{d_i - G_{i1} \cdot m_1}{G_{i2}}$$



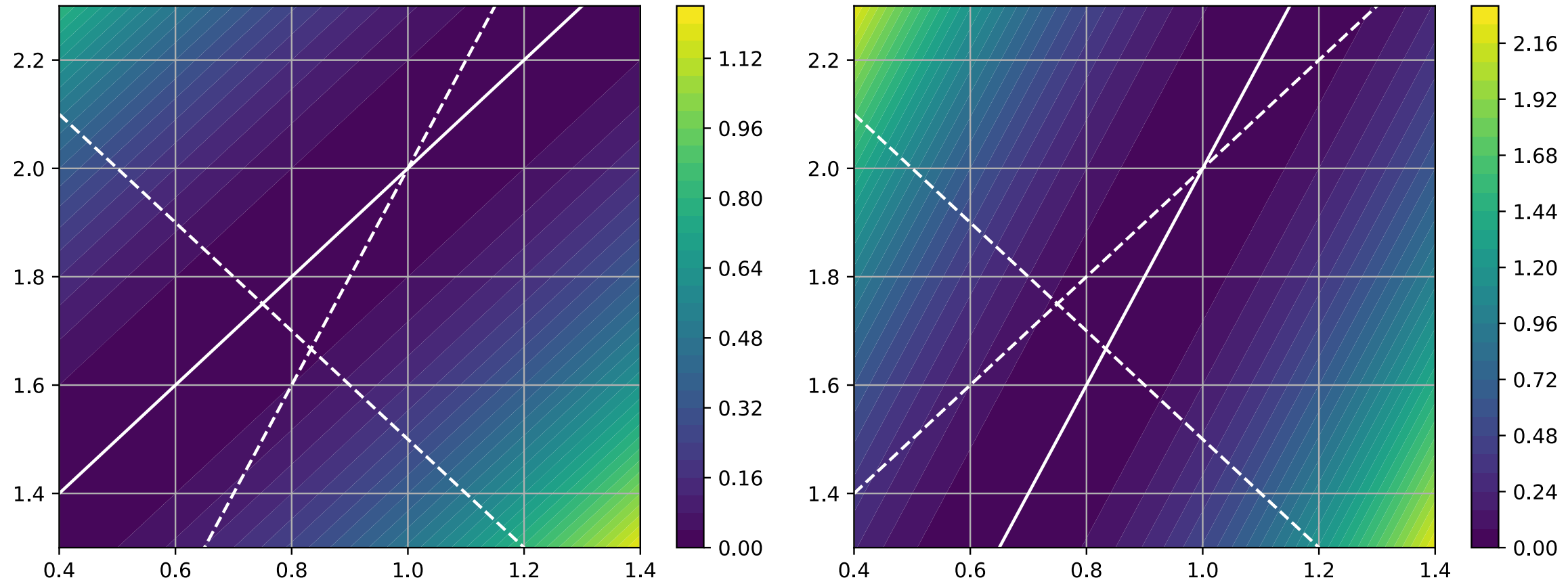
The objective function

We minimize the L2-norm (summed squared distances) of the residual between data and model response:

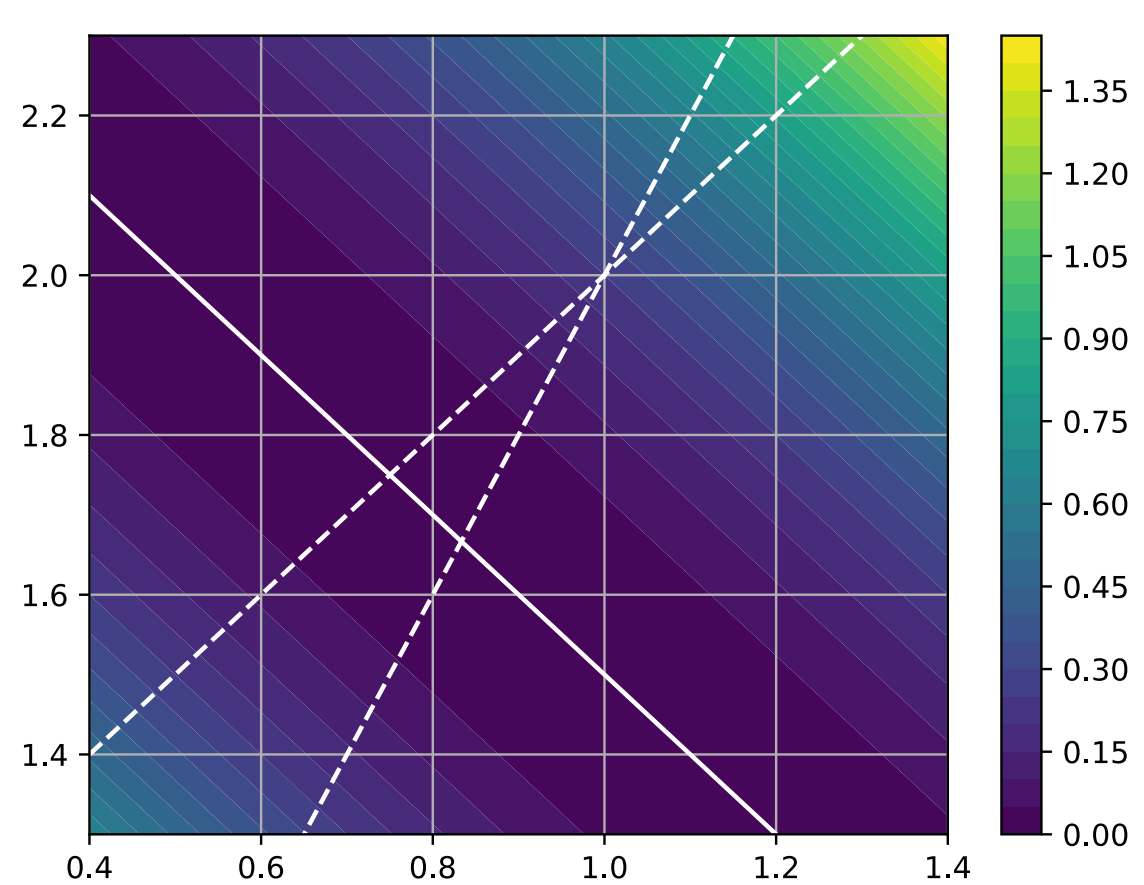
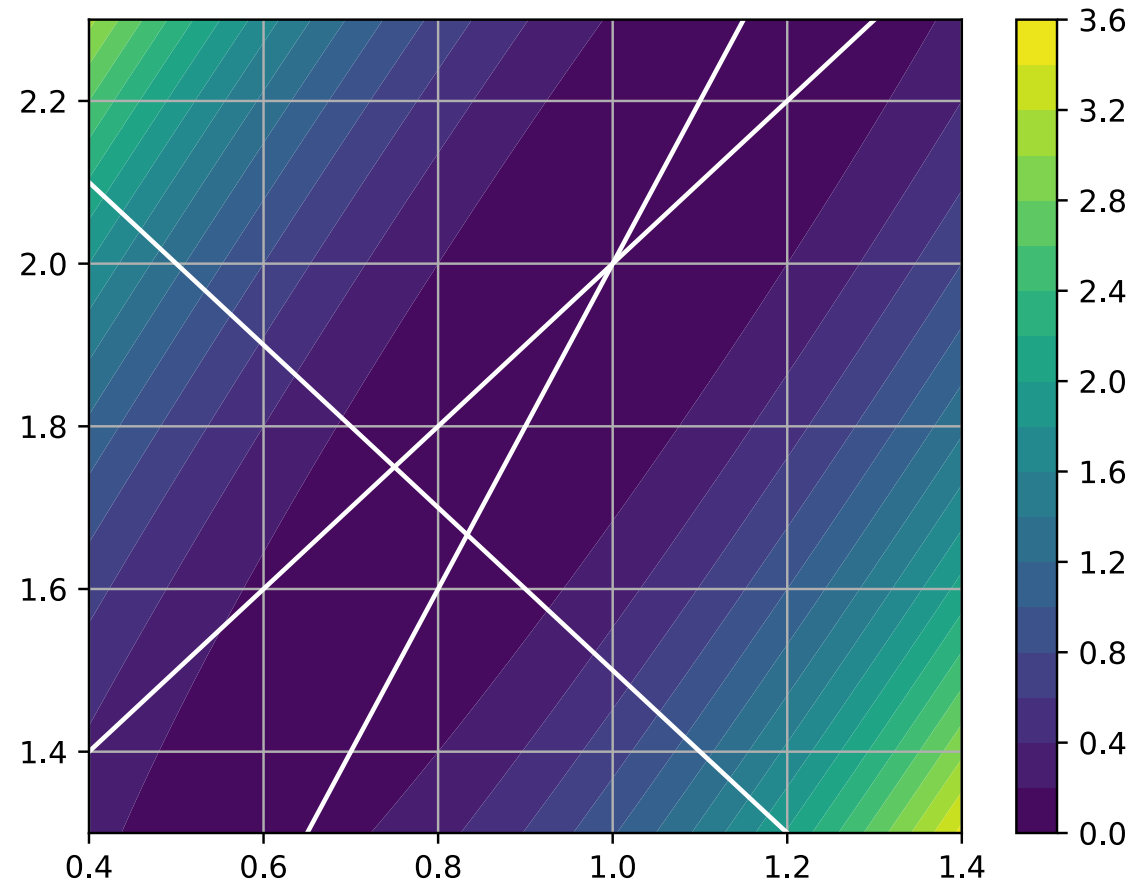
$$\Phi = \|\mathbf{d} - \mathbf{G}\mathbf{m}\|_2^2 = \sum_1^N (\mathbf{d}_i - \mathbf{g}_i(\mathbf{m}))^2 \rightarrow \min$$

Let's compute the objective function for a range of values for m_1 and m_2 (grid search).

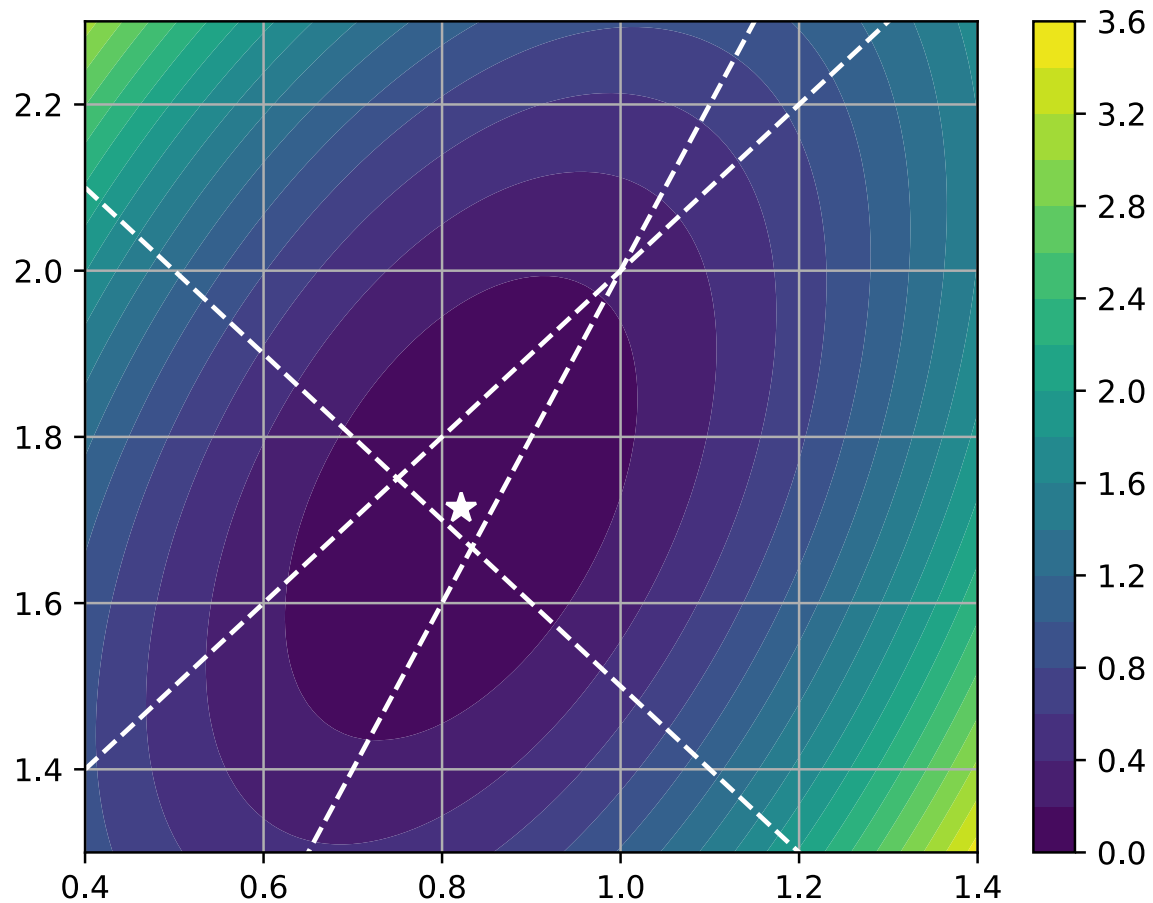
Distance function for the first two data



Distance function of 1st + 2nd and 3rd



Total objective function



- Add all three squared distances.
- ellipsoidal minimum shape
- The minimum pixel (grid spacing of 0.1) lies at (0.82, 1.71).

How can we determine the absolute minimum value?

The method of least squares

Derivation

$$\Phi = \|\mathbf{d} - \mathbf{G}\mathbf{m}\|_2^2 = (\mathbf{d} - \mathbf{G}\mathbf{m})^T (\mathbf{d} - \mathbf{G}\mathbf{m})$$

$$\nabla_m = \left(\frac{\partial}{\partial m_1}, \frac{\partial}{\partial m_2}, \dots, \frac{\partial}{\partial m_M} \right)^T$$

$$\nabla_m \Phi = \nabla_m (\mathbf{G}\mathbf{m} - \mathbf{d})^T (\mathbf{G}\mathbf{m} - \mathbf{d}) = 0$$

$$\nabla_m \Phi = \nabla_m (\mathbf{m}^T \mathbf{G}^T - \mathbf{d}^T) (\mathbf{G}\mathbf{m} - \mathbf{d}) = 0$$

$$\nabla_m \Phi = \nabla_m \mathbf{m}^T \mathbf{G}^T (\mathbf{G}\mathbf{m} - \mathbf{d}) = 0$$

Derivation (cont.)

$$\nabla_m \Phi = \nabla_m \mathbf{m}^T \mathbf{G}^T (\mathbf{G} \mathbf{m} - \mathbf{d}) = 0$$

$$\nabla_m \Phi = \mathbf{G}^T (\mathbf{G} \mathbf{m} - \mathbf{d}) = \mathbf{G}^T \mathbf{G} \mathbf{m} - \mathbf{G}^T \mathbf{d} = 0$$

$$\Rightarrow \quad \mathbf{m} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{d} = \mathbf{G}^\dagger \mathbf{d}$$

$$\text{mit } \mathbf{G}^\dagger = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T$$

(generalized inverse or Pseudo-inverse, in Matlab/Julia $\mathbf{G} \backslash \mathbf{d}$)

The solution $\mathbf{m}_{LS} = \mathbf{G}^\dagger \mathbf{d}$ is the least-squares solution